

Cube Mathematical Domain Model

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Cube Domain Model & Coordinate Geometry

Product: Cubyntra

Organization: Necookie Labs

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1. Domain Standards & Canonical Orientation

Cubyntra adopts standard international World Cube Association (WCA) and Singmaster notation standards.

1.1 Canonical Center Color Scheme (Western Standard)

In canonical orientation:

- Up (U): White (Top)
- Front (F): Green (Front)
- Right (R): Red (Right)
- Down (D): Yellow (Bottom)
- Left (L): Orange (Left)
- Back (B): Blue (Back)

Opposing pairs:

- White $\longleftrightarrow$ Yellow ($U \leftrightarrow D$)
- Green $\longleftrightarrow$ Blue ($F \leftrightarrow B$)
- Red $\longleftrightarrow$ Orange ($R \leftrightarrow L$)

2. 54-Facelet Layout & Indexing

Each face contains 9 stickers indexed row-major from 0 to 8:

```
+---+---+---+
| 0 | 1 | 2 |
+---+---+---+
| 3 | 4 | 5 |   UP (White)
+---+---+---+
| 6 | 7 | 8 |

+---+---+---+---+---+---+---+---+---+---+---+---+
| 0 | 1 | 2 | 0 | 1 | 2 | 0 | 1 | 2 | 0 | 1 | 2 |
+---+---+---+---+---+---+---+---+---+---+---+---+
| 3 | 4 | 5 | 3 | 4 | 5 | 3 | 4 | 5 | 3 | 4 | 5 |
+---+---+---+---+---+---+---+---+---+---+---+---+
| 6 | 7 | 8 | 6 | 7 | 8 | 6 | 7 | 8 | 6 | 7 | 8 |
LEFT (Orange) FRONT (Green) RIGHT (Red) BACK (Blue)
+---+---+---+---+---+---+---+---+---+---+---+---+
| 0 | 1 | 2 |
+---+---+---+
| 3 | 4 | 5 |   DOWN (Yellow)
+---+---+---+
```

| 6 | 7 | 8 |
+---+---+---+

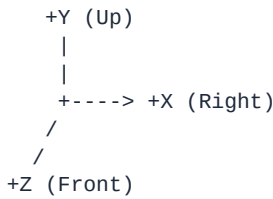
Sticker index 4 on every face is the fixed center piece:

- $U[4] = \text{white}$
- $R[4] = \text{red}$
- $F[4] = \text{green}$
- $D[4] = \text{yellow}$
- $L[4] = \text{orange}$
- $B[4] = \text{blue}$

3. 3D Cubie Coordinate System

The 3D digital twin models the cube as 27 discrete cubies positioned in a right-handed Cartesian coordinate system:

$x, y, z \in \{-1, 0, 1\}$



- +X: Right face (Red)
- X: Left face (Orange)
- +Y: Up face (White)
- Y: Down face (Yellow)
- +Z: Front face (Green)
- Z: Back face (Blue)

Cubie Classification:

- Core Cubie (1 piece): $(0, 0, 0)$ — Internal structural core; no exterior stickers.
- Center Cubies (6 pieces): Exactly one coordinate has magnitude 1 (e.g. $(0, 1, 0)$ to U center).
- Edge Cubies (12 pieces): Exactly two coordinates have magnitude 1 (e.g. $(0, 1, 1)$ to UF edge).
- Corner Cubies (8 pieces): All three coordinates have magnitude 1 (e.g. $(1, 1, 1)$ to UFR corner).

4. Facelet-to-Cubie Spatial Mapping

Face	Coordinate Axis	Facelet Index	Cubie Coordinates (x, y, z)
U	$+Y$ ($y = 1$)	$0 \dots 2$ (Row 0) $3 \dots 5$ (Row 1) $6 \dots 8$ (Row 2)	$(-1, 1, -1), (0, 1, -1), (1, 1, -1)$ $(-1, 1, 0), (0, 1, 0), (1, 1, 0)$ $(-1, 1, 1), (0, 1, 1), (1, 1, 1)$
D	$-Y$ ($y = -1$)	$0 \dots 2$ (Row 0) $3 \dots 5$ (Row 1) $6 \dots 8$ (Row 2)	$(-1, -1, 1), (0, -1, 1), (1, -1, 1)$ $(-1, -1, 0), (0, -1, 0), (1, -1, 0)$ $(-1, -1, -1), (0, -1, -1), (1, -1, -1)$
F	$+Z$ ($z = 1$)	$0 \dots 2$ (Row 0) $3 \dots 5$ (Row 1) $6 \dots 8$ (Row 2)	$(-1, 1, 1), (0, 1, 1), (1, 1, 1)$ $(-1, 0, 1), (0, 0, 1), (1, 0, 1)$ $(-1, -1, 1), (0, -1, 1), (1, -1, 1)$
B	$-Z$ ($z = -1$)	$0 \dots 2$ (Row 0) $3 \dots 5$ (Row 1) $6 \dots 8$ (Row 2)	$(1, 1, -1), (0, 1, -1), (-1, 1, -1)$ $(1, 0, -1), (0, 0, -1), (-1, 0, -1)$ $(1, -1, -1), (0, -1, -1), (-1, -1, -1)$
R	$+X$ ($x = 1$)	$0 \dots 2$ (Row 0) $3 \dots 5$ (Row 1) $6 \dots 8$ (Row 2)	$(1, 1, 1), (1, 1, 0), (1, 1, -1)$ $(1, 0, 1), (1, 0, 0), (1, 0, -1)$ $(1, -1, 1), (1, -1, 0), (1, -1, -1)$
L	$-X$ ($x = -1$)	$0 \dots 2$ (Row 0) $3 \dots 5$ (Row 1) $6 \dots 8$ (Row 2)	$(-1, 1, -1), (-1, 1, 0), (-1, 1, 1)$ $(-1, 0, -1), (-1, 0, 0), (-1, 0, 1)$ $(-1, -1, -1), (-1, -1, 0), (-1, -1, 1)$

5. Face Rotation Permutations

When a face rotates 90° Clockwise (CW), two transformations occur simultaneously:

- The 3x3 face matrix rotates 90° CW:
$$\begin{bmatrix} 0 & 1 & 2 \\ 3 & 4 & 5 \\ 6 & 7 & 8 \end{bmatrix} \rightarrow \begin{bmatrix} 6 & 3 & 0 \\ 7 & 4 & 1 \\ 8 & 5 & 2 \end{bmatrix}$$
- The 4 adjacent face strips cycle 3 stickers each:

Example: U Turn (Clockwise around +Y)

- $F[0, 1, 2] \rightarrow L[0, 1, 2]$
- $L[0, 1, 2] \rightarrow B[0, 1, 2]$
- $B[0, 1, 2] \rightarrow R[0, 1, 2]$
- $R[0, 1, 2] \rightarrow F[0, 1, 2]$

Example: R Turn (Clockwise around +X)

- $U[2, 5, 8] \rightarrow B[6, 3, 0]$ (reversed)
- $B[6, 3, 0] \rightarrow D[2, 5, 8]$
- $D[2, 5, 8] \rightarrow F[2, 5, 8]$
- $F[2, 5, 8] \rightarrow U[2, 5, 8]$

6. Mathematical Invariants & Verification

The Rubik's Cube group has order:

$$|G| = \frac{8! \times 3^8 \times 12! \times 2^{12}}{2 \times 3 \times 2} = 43,252,003,274,489,856,000$$

Cubyntra validates all three invariant constraints:

- Corner Orientation Parity: Each corner has a twist in $\{0, 1, 2\}$. The total twist must sum to zero mod 3:
$$\sum_{i=1}^8 \text{twist}_i \equiv 0 \pmod{3}$$
- Edge Orientation Parity: Each edge has a flip in $\{0, 1\}$. The total flips must sum to zero mod 2:
$$\sum_{i=1}^{12} \text{flip}_i \equiv 0 \pmod{2}$$
- Permutation Parity: The permutation sign of the 8 corners must match the permutation sign of the 12 edges:
$$\text{sgn}(\sigma_{\text{corners}}) = \text{sgn}(\sigma_{\text{edges}})$$

Any state violating these invariants is physically unsolvable without disassembly.
